

GIFT: Glaucoma Imaging of Future Trajectories

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Abstract

We developed GIFT (Glaucoma Imaging of Future Trajectories), an open-source research tool that generates future retinal images from one photograph at times up to five years ahead. In 1,636 test pairs from 49 patients, generated images achieved 0.9645 global similarity to processed real follow-up images. This supports single-photograph image generation for clinician assessment; demonstrating accurate glaucoma prediction and equivalence to multi-visit methods requires further evaluation.

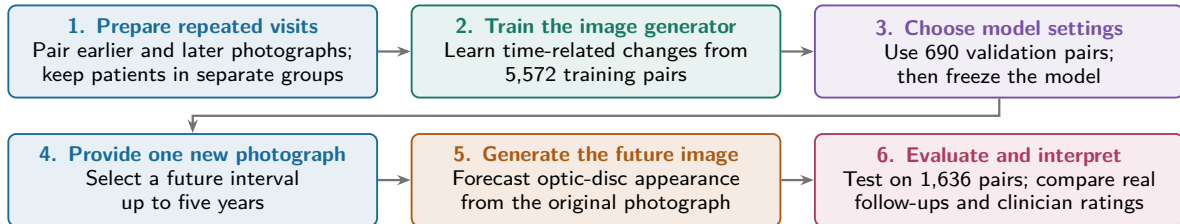
Introduction

Glaucoma damages the optic nerve, which carries visual information to the brain. Retinal photographs show its visible head, the optic disc, and central depression, the optic cup; changes in these structures help clinicians assess disease¹. We wanted to help clinicians see possible future changes before they occur. Earlier work predicts glaucoma from two visits² or generates future images from several visits³. We asked whether one photograph could provide useful forecasts for studying glaucoma onset and worsening, with less patient history.

Methods

We retrospectively studied Sequential Fundus Images for Glaucoma Forecast (SIGF)⁴: 3,684 photographs from 405 eyes of 262 patients. We assigned each patient’s images to one group, using 5,572 earlier–later pairs for training, 690 for choosing settings, and 1,636 for testing. PAPILA, a separate dataset, supplied expert disc and cup outlines for our measurement tool⁵.

GIFT learns to generate images by removing added noise⁶, guided by time and photographic vessel, nerve-fiber, and brightness patterns. Each forecast starts from the same real photograph. We measured image error and similarity against processed real follow-ups and an unchanged starting-image reference. Clinician assessment concerns resemblance and the amount of visible change.



Results

Generated images closely resembled processed follow-ups: global similarity was 0.9645 (1 means identical), and average pixel error was 0.044255 on a 0–1 scale. The unchanged image gave nearly identical scores (Table 2); GIFT’s error was slightly lower near three and five years. Two example eyes showed mean automatic cup–outline overlap of 0.869.

Clinician-results template; complete after review: Dr. Gee reviewed [n] final follow-ups during [dates]. Mean similarity was [score]/5; [n/N; %] showed about the same amount of change as the real image (Table 4).

Discussion

GIFT is an open-source research tool for clinicians to explore glaucoma-related appearance up to five years ahead from one photograph. Its contribution is the reduced input requirement; equal performance to multi-visit methods remains unproven. Image resemblance also needs validation against glaucoma onset and worsening. The approach could extend to other eye diseases with disease-specific longitudinal training and evaluation. Sitar Eswar led the model implementation and analysis.

Study	Input history	What it produces	Generated versus real image evidence
Lin and colleagues [2]	2 visits of the same eye	A future-glaucoma prediction	No generated images to compare.
Zhang and colleagues [3]	Several earlier visits of the same eye	A glaucoma prediction and a future retinal image	Two visual examples; no numerical image-similarity score reported.
GIFT (ours)	1 image; future time	A future optic-disc image for clinician interpretation	Similarity: 0.9645. Pixel error: 0.044255. Tested on 1,636 image pairs.

Table 1: Earlier approaches and GIFT. Zhang generates future images; Lin predicts glaucoma without image generation. Disease-classification scores cannot establish equal image quality across studies.

Time ahead	Pairs	GIFT error	Unchanged image error	Error reduction	GIFT similarity	Unchanged image similarity
All	1,636	0.044255	0.044198	−0.13%	0.964467	0.964469
~1 year	464	0.040055	0.039421	−1.61%	0.968325	0.968577
~3 years	753	0.043956	0.044026	+0.16%	0.964745	0.964604
~5 years	419	0.049445	0.049795	+0.70%	0.959694	0.959676

Table 2: Results from 49 test patients. Error is mean absolute error (lower is better); similarity is the global structural similarity index measure (1 is identical). Positive reduction favors GIFT. Scores use aligned, color-matched 128×128 targets with changes clipped to ± 0.18 . Patient-averaged error difference: 0.000165; 95% confidence interval from patient resampling: $[-0.000147, 0.000513]$. Time groups: ≤ 1.5 , > 1.5 – 3.5 , and > 3.5 –5 years.

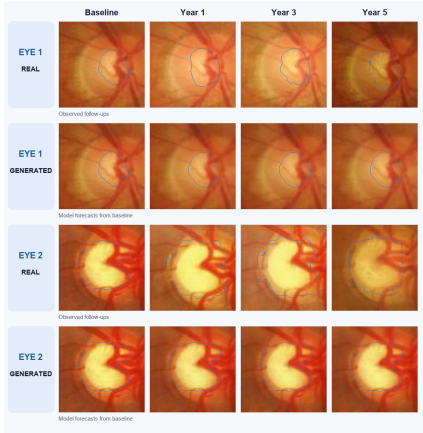


Figure 1: Two example eyes. Baseline is the starting photograph; generated is the forecast. Columns show visits nearest Years 1, 3, and 5. Blue lines mark automatic cup outlines.

Measure in example eyes	Year 1	Year 3	Year 5
Cup overlap (Dice score)	0.860	0.869	0.879
Cup-to-disc height ratio error	0.112	0.094	0.154
Vessel-pattern agreement	0.456	0.589	0.369
Nerve-fiber pattern agreement	0.641	0.765	0.616

Table 3: Automatic averages for the two example eyes. Dice measures cup-outline overlap; 1 is perfect. Height ratio divides cup height by disc height. Pattern agreement is correlation. These are photographic measures, not clinician ratings.

Clinician-results template	Result
Final follow-ups reviewed (number)	_____
Mean similarity rating (1–5)	_____
About the same change (number/total; %)	_____
Review dates	_____

Table 4: Draft results skeleton. Similarity ranges from 1 (very different) to 5 (very similar). Use assessable cases as the denominator; report unable-to-assess counts separately. Complete blanks only after review.

References

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